AIUB | FST | CSC 4233 Natural Language Processing **Group:** G07 | [Section: B] | Mid-Term Project Proposal (M2)

**Sentiment Classification of IMDB Movie Reviews Using Classical NLP Techniques**

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## 1. Problem Statement and Use Case

We will build an end-to-end binary text classifier that labels an English movie review as positive or negative. Automatic sentiment analysis helps streaming services and review sites summarize user opinion, rank titles, and detect deceptive ratings without manually reading thousands of reviews. Our goal is to demonstrate, using only Weeks 1–5 techniques (no transformers), which preprocessing, representation, and classifier choices work best for this task.

## 2. Dataset

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| **Source** | IMDB Large Movie Review Dataset (Maas et al., ACL 2011), Stanford AI Lab; loaded in the notebook from a public link (Stanford/Hugging Face imdb mirror), with a backup copy in the ZIP. |
| **Size / classes** | 50,000 labelled reviews (25,000 train + 25,000 test), 2 classes (positive, negative). Far above the 2,000-document minimum. |
| **Class balance** | Perfectly balanced (50% / 50%) in the original data, so the data is **not** imbalanced. Macro-F1 remains our primary metric. |
| **Sampling** | The dataset is large, so we will draw a **stratified random sample of 10,000 reviews (5,000 per class)** with a fixed seed and report this in the report. Split: stratified 70/15/15 = 7,000 train / 1,500 validation / 1,500 test. |
| **Licence** | Publicly released by the authors for research use; we will cite Maas et al. (2011) and confirm academic-use terms on the dataset page before the final submission. |

## 3. Planned Pipeline and Models

- **EDA:** Class distribution, review-length statistics, and top words per class before and after preprocessing — at least three figures, each with a one-line interpretation.
- **Preprocessing (reusable function):** regex cleaning (HTML tags such as <br />, URLs, numbers, emojis), tokenization, lower-casing, punctuation and stop-word removal, and stemming/lemmatization. Since negation words (not, no, never) matter for sentiment, we will test keeping them. **Ablation:** at least two variants compared on macro-F1 — with vs. without stop-word removal, and stemming vs. lemmatization.
- **Representations:** Bag of Words (count vectors); TF-IDF with unigram vs. unigram + bigram; averaged word embeddings (Word2Vec trained on our corpus or pretrained GloVe/FastText).
- **Classifiers:** a majority-class baseline; Multinomial Naive Bayes on BoW and TF-IDF (required); Logistic Regression (or Linear SVM) on all three representations, using Logistic Regression for embeddings. At least one hyperparameter per model will be tuned on the validation set (e.g., NB alpha, C, max_features, n-gram range).
- **Evaluation:** stratified split with a fixed seed; vectorisers fitted on training data only; test set used once. One comparison table with accuracy, macro-precision, macro-recall and macro-F1 per model, plus a confusion matrix for the best model.
- **Error analysis:** top-10 informative words per class from the Naive Bayes log-probability ratio, and at least 20 misclassified test reviews grouped by error type (sarcasm, negation, mixed opinions, label noise).
- **Deliverables:** a Colab-ready .ipynb (runs top to bottom, fixed seeds, pip cell at the top), a 4–6 page IEEE-format report, a contribution statement, and an AI-use statement.
- **Scope note:** No transformer-based models (BERT, GPT, etc.) will be used in this project.

## 4. Risks and Mitigation

- **Sarcasm, negation and long reviews** may confuse bag-of-words models → use bigrams, keep negation words, and discuss these cases in error analysis.
- **Long training time for embeddings on Colab** → use the 10,000-review sample and a small Word2Vec vector size, or pretrained vectors.
- **HTML noise and duplicate reviews** → clean with regex and check for duplicates across splits to avoid data leakage.

## 5. Planned Roles (equal responsibility; every member can explain any cell in the viva).

| Member | Planned role |
|---|---|
| Sadiya Mohon Auti (23-52092-2) | Overall coordination and submissions; preprocessing pipeline and ablation study; classifiers, hyperparameter tuning and evaluation; EDA and report writing, report integration and final review. |
| Tasnim Tabassum Mim (23-50971-1) | Dataset loading and stratified sampling; figures; BoW/TF-IDF/embedding representations; error analysis; notebook documentation (dataset, results sections). |

**Reference:** [1] A. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, "Learning Word Vectors for Sentiment Analysis," *ACL Anthology*, pp. 142–150, June 2011. [Online]. Available: https://aclanthology.org/P11-1015/